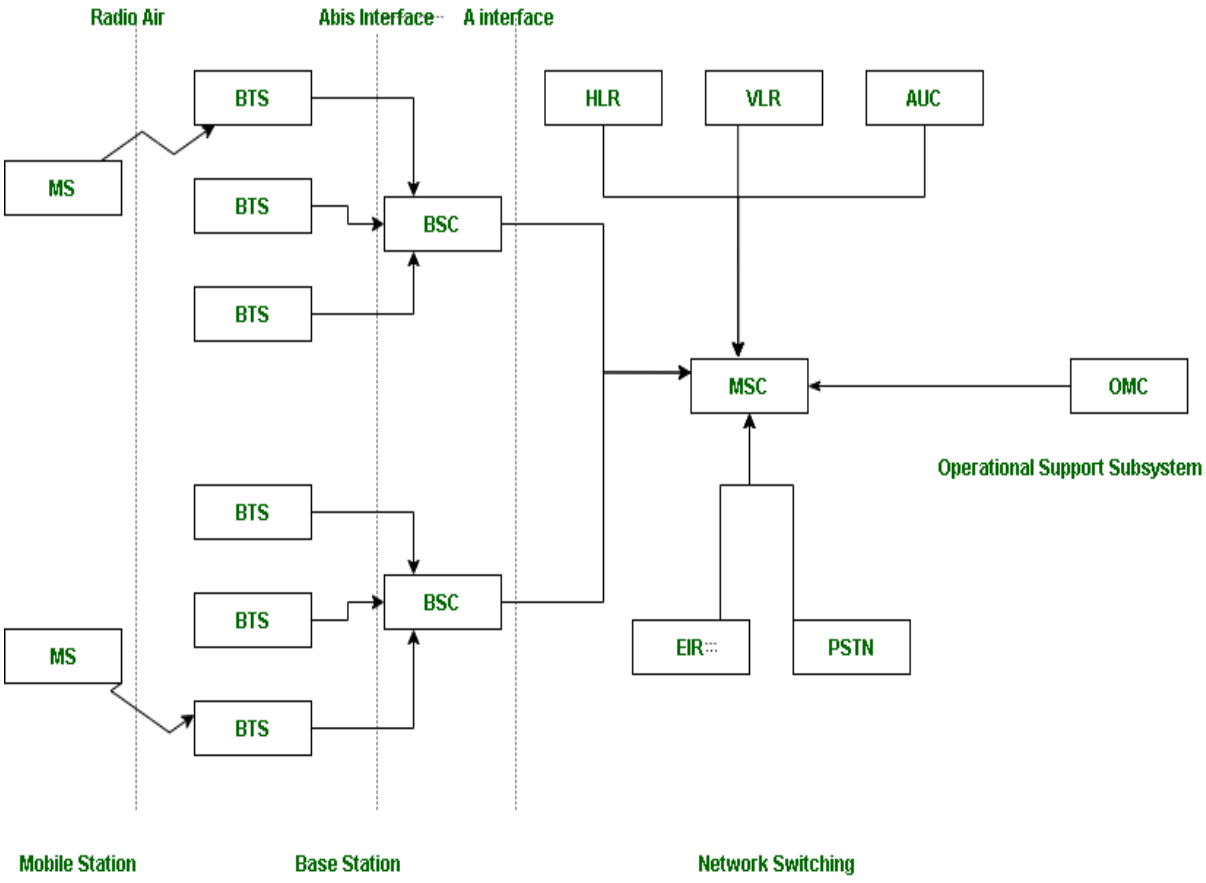
UNIT -2 Important Questions

1. **Explain the GSM architecture with neat sketch**

GSM:

GSM stands for Global System for Mobile Communication. It is a digital mobile network commonly utilized by mobile phone users around the world. It is the most popular of the three digital wireless telephony systems (TDMA, GSM, and CDMA).

Architecture of GSM



MS- M0bile station
BTS- Base Station
BSC- Base Station Controller
MSC- Mobile Switching Center
HLR- Home Location Register
VLR- visitor location register
AUC- authentication center
OMC- Operation Maintenance Center
EIR- Equipment Identity Register
PSTN- Public Switched Telephone Network,

GSM is nothing but a larger system which is divided into further 3 subsystems.

1. **BSS : BSS stands for Base Station Subsystem.**

- **BSS handles traffic and signaling between a mobile phone and the network switching subsystem.**
- **BSS having two components BTS and BSC.**

2. NSS : NSS stands for Network and Switching Subsystem. NSS is the core network of GSM. That carried out call and mobility management functions for mobile phone present in network. NSS have different components like VLR, HLR and EIR.Visitor Locator Register, Home Location Register, Equipment Identity Register

3. OSS : OSS stands for Operating Subsystem. OSS is a functional entity which the network operator monitor and control the system. OMC(operation Maintenance center) is the part of OSS. Purpose of OSS is to offer the customer cost-effective support for all GSM related maintenance services.

1. **MS : MS stands for Mobile System.**

- **MS comprises user equipment and software needed for communication with a mobile network.**
- **Mobile Station (MS) = Mobile Equipment(ME) + Subscriber Identity Module (SIM).**

2. BTS : BTS stands for Base Transceiver Station which facilitates wireless communication between user equipment and a network.

- **Every tower has BTS.**

3. BSC : BSC stands for Base Station Controller.

- **BSC has multiple BTS.**
- **You can consider the BSC as a local exchange of your area which has multiple towers and multiple towers have BTS.**

4. MSC : MSC stands for Mobile Switching Center.

- **MSC is associated with communication switching functions such as call setup, call release and routing.**
- **Call tracing, call forwarding all functions are performed at the MSC level.**
- **MSC is having further components like VLR, HLR, AUC, EIR and PSTN.**

VLR : VLR stands for Visitor Location Register.

- **VLR is a database which contains the exact location of all mobile subscribers currently present in the service area of MSC. If you are going from one state to another state then your entry is marked into the database of VLR.**

HLR : HLR stands for Home Location Register.

- **HLR is a database containing pertinent data regarding subscribers authorized to use a GSM network..**

- **If you purchase SIM card from in the HLR.**
- **HLR is like a home which contains all data like your ID proof, which plan you are taking, which caller tune you are using etc.**

OMC : OMC stands for Operation Maintenance Center.

- **OMC monitor and maintain the performance of each MS, BSC and MSC within a GSM system.**

AUC : AUC stands for Authentication Center. AUC authenticates the mobile subscriber that wants to connect in the network.

EIR : EIR stands for Equipment Identity Register.

- **EIR is a database that keeps the record of all allowed or banned in the network. Store the IMEI no of the subscriber.**

PSTN : PSTN stands for Public Switched Telephone Network.

- **PSTN connects with MSC.**

PSTN originally a network of fixed line analog telephone systems

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2. Discuss about different types of handover

Handover:

- One of the key elements of a mobile phone or cellular telecommunications system, is that the system is split into many small cells to provide good frequency re-use and coverage.
- However as the mobile moves out of one cell to another it must be possible to retain the connection.
- The process by which this occurs is known as handover or handoff.

Types of GSM handover

Within the GSM system there are four types of handover that can be performed for GSM only systems:

- **Intra-BTS handover:** This form of GSM handover occurs if it is required to change the frequency or slot being used by a mobile because of interference, or other reasons. In this form of GSM handover, the mobile remains attached to the same base station transceiver, but changes the channel or slot.
- **Inter-BTS Intra BSC handover:** This for of GSM handover or GSM handoff occurs when the mobile moves out of the coverage area of one BTS but into another controlled by the same BSC. In this instance the BSC is able to perform the handover and it assigns a new channel and slot to the mobile, before releasing the old BTS from communicating with the mobile.

- **Inter-BSC handover:** When the mobile moves out of the range of cells controlled by one BSC, a more involved form of handover has to be performed, handing over not only from one BTS to another but one BSC to another. For this the handover is controlled by the MSC.
- **Inter-MSC handover:** This form of handover occurs when changing between networks. The two MSCs involved negotiate to control the handover.

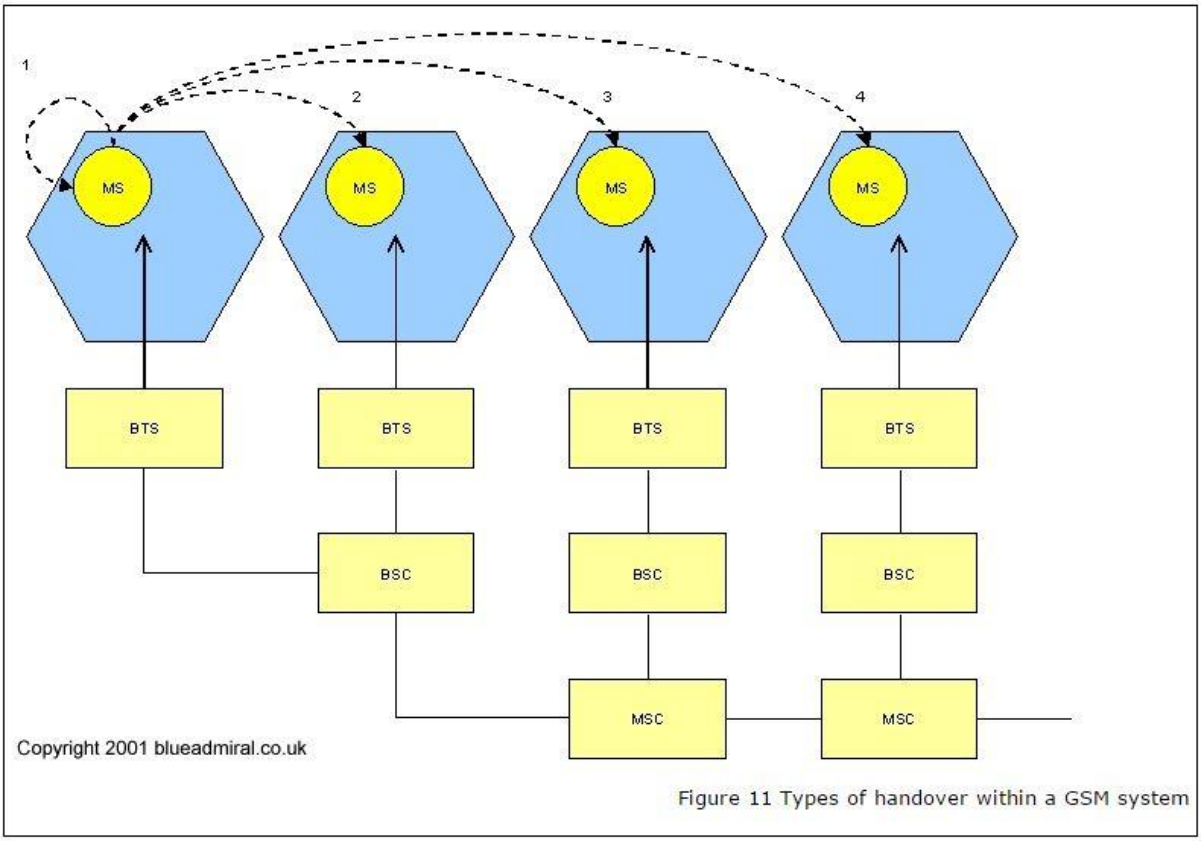


Figure 11 Types of handover within a GSM system

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3. Compare and Contrast HSPA+, HSUPA and HSDPA+

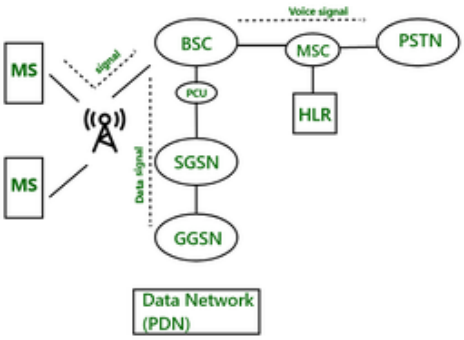
Feature	HSDPA+	HSUPA	HSPA+
Full form	High-Speed Downlink Packet Access Plus	High-Speed Uplink Packet Access	Evolved High-Speed Packet Access
Main purpose	Faster **download**	Faster **upload**	Faster **download and upload**
Direction	Downlink	Uplink	Both
Technology	Enhancement of HSDPA	Enhancement of UMTS uplink	Evolution of HSPA

Feature	HSDPA+	HSUPA	HSPA+
Typical peak speed	Up to ~42 Mbps or more, depending on implementation	Up to ~5.76 Mbps	Up to ~42 Mbps downlink and ~11.5 Mbps uplink in common deployments
Main users	Users downloading data	Users uploading data	Users requiring high-speed two-way data
Key benefit	Improved download performance	Improved upload performance	Higher overall throughput and lower latency
Relation	Part of HSPA family	Part of HSPA family	Advanced version of HSPA

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4. **Explain the GPRS architecture with neat sketch**

- **GPRS stands for General Packet Radio Service.**
- **It is the enhanced version of GSM architecture service.**
- **In GSM architecture we can only transport the voice signal.**
- **But if signal consists of the data then only GSM architecture cannot use.**
- **For that there are two more software components are used, SGSN (Serving GPRS supporting Node) and GGSN (Gateway GPRS supporting Node).**
- **In GSM architecture there is one component called BSC. But in GPRS there is one component is added to BSC called PCU(packet Control unit).**
- **PCU stands for Packet Control Unit. If signal comes to BSC and that signal contains data, then PCU routes to the SGSN. Interface is used between BSC and PCU is FRI interface.**
- **After signal comes to SGSN, it delivers the data packet to the GGSN. GGSN routes the data packet to the data network (PDN- Predefined Data Network).**



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5. **Explain the architecture of UMTS with suitable diagram.**

- **UMTS (Universal Mobile Telecommunications System) is a 3G (third-generation) mobile communication technology used in mobile computing to provide high-speed wireless data**

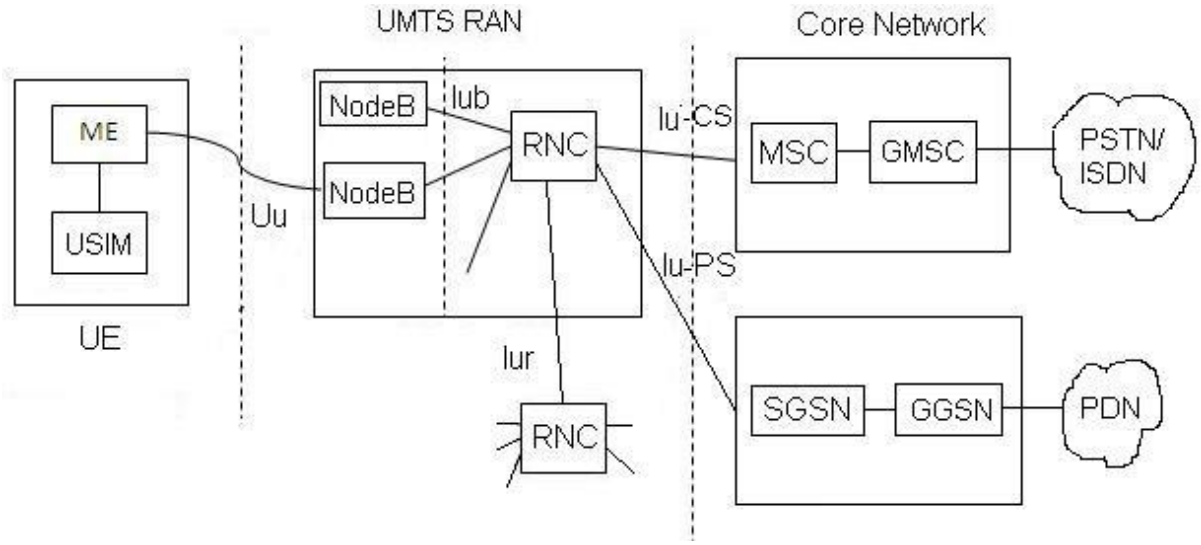
and voice services over cellular networks. It contain three part User Equipment, Access Network, Core Network.

- The 3GPP (3rd Generation Partnership Project) developed and maintains UMTS, which seeks to provide faster data transfer speeds and expanded capabilities than its predecessor, GSM (2G

3G UMTS employments a completely diverse radio interface based around the utilize of Coordinate Grouping Spread Range as CDMA or Code Division Multiple Access

Abbreviation	Full Form	Description
UE	User Equipment	Mobile device used by the subscriber
ME	Mobile Equipment	The actual mobile device (phone/tablet)
USIM	Universal Subscriber Identity Module	Smart card containing subscriber identity and security information
UTRAN	UMTS Terrestrial Radio Access Network	Radio access part of UMTS consisting of Node B and RNC
Node B	Node Base Station	UMTS base station that communicates directly with mobile devices
RNC	Radio Network Controller	Controls Node Bs and manages radio resources
CN	Core Network	Main network responsible for switching, routing, and service control
MSC	Mobile Switching Centre	Handles circuit-switched voice calls and mobility management
GMSC	Gateway Mobile Switching Centre	Gateway between UMTS network and external telephone networks
SGSN	Serving GPRS Support Node	Handles packet data delivery and user mobility
GGSN	Gateway GPRS Support Node	Connects UMTS packet network to external packet data networks
PSTN	Public Switched Telephone Network	Traditional telephone network

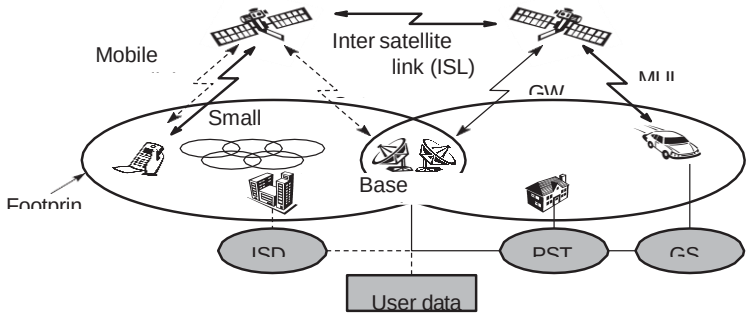
ISDN	**Integrated Services Digital Network**	**Digital telephone communication network**
PDN	**Packet Data Network**	**External packet-based network such as the Internet**



6. **Discuss about satellite systems and its applications**

A satellite system is a communication network in which a satellite acts as a relay station. It receives signals from a ground station or mobile device, amplifies them, and retransmits them to another ground station or user.





Components of a Satellite System

- **Satellite – Receives and retransmits signals.**
- **Ground Station (Earth Station) – Sends and receives signals to/from the satellite.**
- **Mobile User Terminal – Devices such as satellite phones, ships, aircraft, or remote communication terminals.**

Working of a Satellite System

- **A mobile device sends a signal to the satellite (uplink).**
- **The satellite receives and processes the signal.**
- **The satellite retransmits the signal back to Earth (downlink) to another user or ground station.**

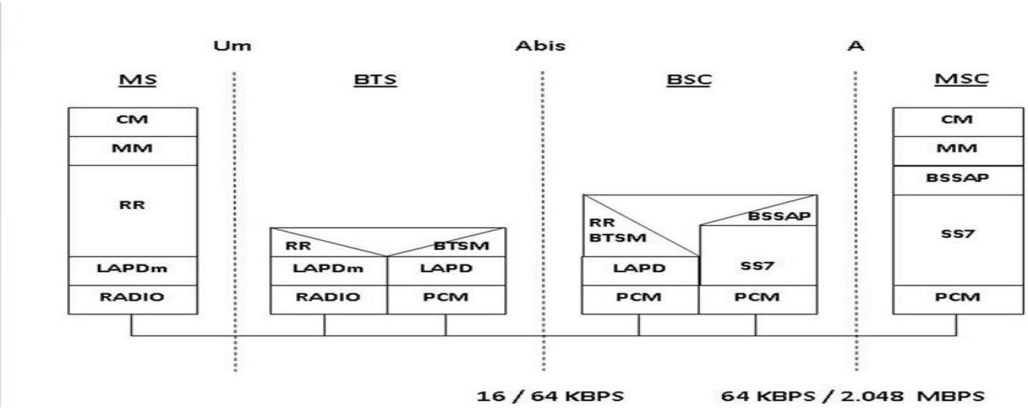
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7. Name the layers of GSM protocol architecture. Explain with diagram.

GSM :

GSM stands for Global System for Mobile Communication. It is a digital mobile network commonly utilized by mobile phone users around the world. It is the most popular of the three digital wireless telephony systems (TDMA, GSM, and CDMA).

GSM Protocol Architecture:



The layered model of the GSM architecture integrates and links the peer-to-peer communications between two different systems.

The underlying layers satisfy the services of the upper-layer protocols.

Notifications are passed from layer to layer to ensure that the information has been properly formatted, transmitted, and received.

MS Protocols:

The signaling protocol in GSM is structured into three general layers, depending on the interface.

Layer 1: **The physical layer, which uses the channel structures over the air interface. Radio interface.**

Layer 2: **The data-link layer. Across the Um interface, the data-link layer is a modified version of the Link access protocol for the D channel (LAP-D) protocol used in ISDN, called Link access protocol on the Dm channel (LAP-Dm). It is used for flow control error control and authentication.**

Layer 3: **The third layer of the GSM signaling protocol is divided into three sublayers:**

1. **Radio Resource management (RR)**
2. **Mobility Management (MM) and**
3. **Connection Management (CM).**

The MS to BTS Protocols:

- **The RR layer oversees the establishment of a link, both radio and fixed, between the MS and the MSC.**
- **The main functional components involved are the MS, the BSS, and the MSC.**
- **The RR layer is concerned with the management of an RR-session, which is the time that a mobile is in dedicated mode, as well as the configuration of radio channels, including the allocation of dedicated channels.**
- **The MM layer is built on top of the RR layer and handles the functions that arise from the mobility of the subscriber, as well as the authentication and security aspects.**
- **Location management is concerned with the procedures that enable the system to know the current location of a powered-on MS so that incoming call routing can be completed.**
- **The CM layer is responsible for CC, supplementary service management, and Short Message Service (SMS) management.**

BSC Protocols:

- **After the information is passed from the BTS to the BSC, a different set of interfaces is used.**
- **The Abis interface is used between the BTS and BSC.**
- **At this level, the radio resources at the lower portion of Layer 3 are changed from the RR to the Base Transceiver Station Management (BTSM). The BTS management layer is a relay function at the BTS to the BSC.**
- **The RR protocols are responsible for the allocation and reallocation of traffic channels between the MS and the BTS.**

- **These services include controlling the initial access to the system, paging for MT calls, the handover of calls between cell sites, power control, and call termination.**
- **From the BSC, the relay is using SS7 protocols so the MTP 1-3 is used as the underlying architecture, and the BSS mobile application part or the direct application part is used to communicate from the BSC to the MSC.**

MSC Protocols:

- **At the MSC, the information is mapped across the A interface to the MTP Layers 1 through 3 from the BSC.**
- **Here, the equivalent set of radio resources is called the BSS MAP. The BSS MAP/DTAP and the MM and CM are at the upper layers of Layer 3 protocols.**
- **This completes the relay process. Through the control-signaling network, the MSCs interact to locate and connect to users throughout the network.**

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8. Give reasons for a handover in GSM and the problems associated with it .Discuss the typical steps for handover are and what types of handover can occur?

Handover (or handoff) is the process of transferring an ongoing call or data connection from one GSM cell/base station to another without interrupting the communication.

Reasons for handover in GSM

1. **Poor signal strength:** When the received signal from the serving BTS becomes too weak, the call is transferred to a cell with a stronger signal.
2. **Poor signal quality:** High interference or a high bit-error rate (BER) may require handover.
3. **Moving between cells:** When a mobile station moves from one cell to another, handover maintains the connection.
4. **Traffic/load balancing:** A mobile may be handed over to another cell when the serving cell is congested.
5. **Interference reduction:** Handover can move the mobile to a frequency/cell with less interference.
6. **Maintaining connection:** Handover prevents a call from being dropped as the mobile moves away from the serving BTS.

Problems associated with handover

Call dropping: If a suitable target cell cannot be found, the ongoing call may be terminated.

Handover delay: The connection can be briefly interrupted during the transfer.

Signalling overhead: Handover requires communication between the mobile station, BTS, BSC and possibly MSC.

Interference: Poorly planned frequency allocation can cause interference during handover.

Resource unavailability: The target cell may not have a free traffic channel.

Typical steps involved in GSM handover

1. **Measurement:** The mobile station continuously measures the signal strength and quality of the serving cell and neighboring cells.
2. **Measurement reporting:** The MS sends measurement reports to the serving BTS/BSC.
3. **Channel allocation:** A traffic channel is reserved in the target cell.
4. **Handover command:** The network sends a handover command to the mobile station.
5. **Handover completion:** The network switches the connection to the new cell and releases the old channel.

Types of Handover

1. Intra-cell handover
The mobile remains within the same cell but changes its radio channel, usually because of interference or poor channel quality.

2. Inter-cell handover
The mobile moves from one cell to another. This is the most common type.

3. Intra-BSC handover: Both the old and new cells are controlled by the same BSC.

4. **Inter-BSC handover:** The old and new cells are controlled by different BSCs but generally belong to the same MSC.
5. **Inter-MSC handover:** The mobile moves between cells controlled by different MSCs.

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9. **Show the steps involved in a mobile terminated call with diagram (a station calling a mobile station) in GSM**

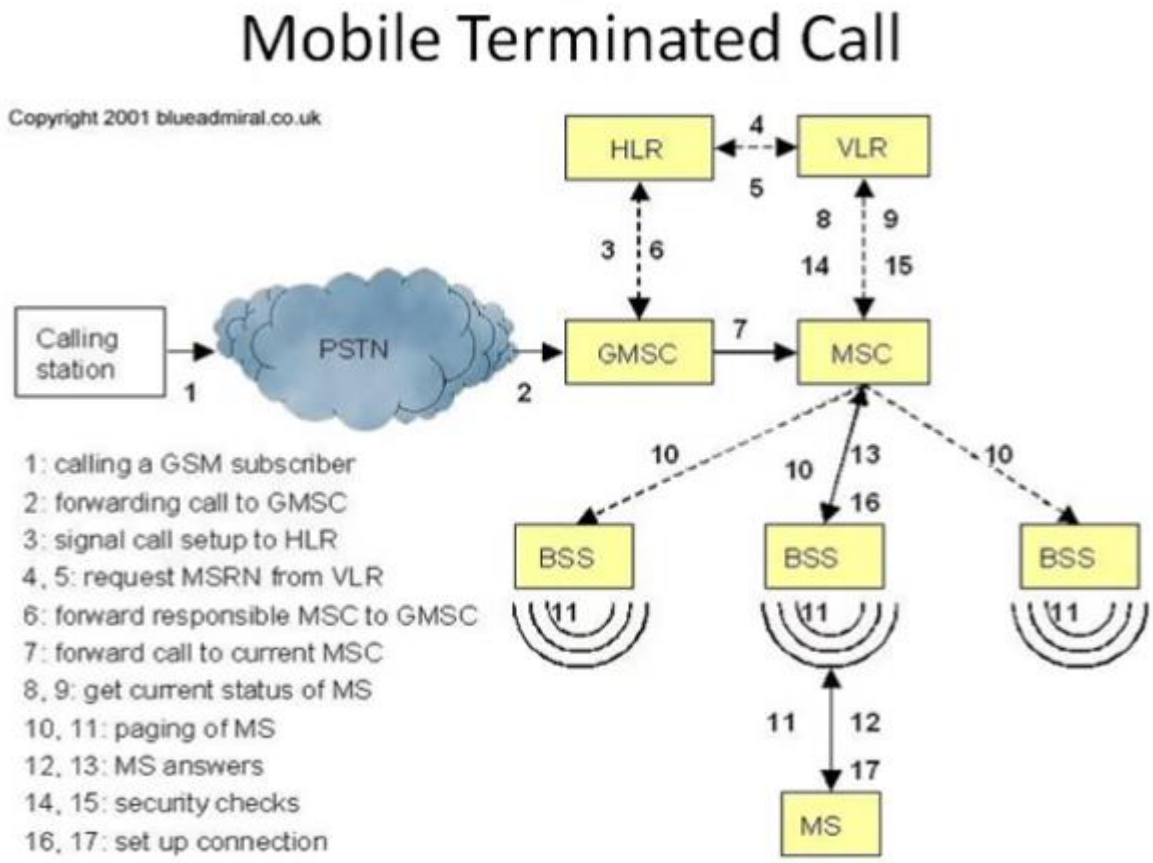
MOBILE TERMINATED CALL:

Basic steps needed to connect the calling station with the mobile user are:

- **Step 1: a user dials the phone number of a GSM subscriber.**
- **Step 2: The fixed network (PSTN) notices that the number belongs to a user in the GSM network and forwards the call setup to the Gateway MSC.**
- **Step 3: The GMSC identifies the HLR for the subscriber and signals the call setup to the HLR.**
- **Step 4: The HLR now checks whether the number exists and whether the user has subscribed to the requested services, and requests an MSRN from the current VLR.**
- **Step 5:MSRN is received**

Step 6: the HLR determine the MSC responsible for the MS and forwards this information to the GMSC.

- **Step 7: The GMSC now forward the call setup request to the MSC indicated.**
- **Step 8: MSC requests the current status of the MS from the VLR.**
- **Step 9&10: If MS is available MSC initiates paging in all cells.**
- **Step11: BTSs of all BSSs transmit this paging signal to MS If the MS answers (12 and 13), the VLR perform security checks (encryption). The VLR signals to the MSC to set up a connection to the MS (steps 15 to 17).**



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10. Give reasons for a handover in GSM and the problems Associated with it. Discuss

Reasons for handover in GSM

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4. **Interference:** Poorly planned frequency allocation can cause interference during handover.
5. **Resource unavailability:** The target cell may not have a free traffic channel.

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11. Explain the security services of GSM?

GSM stands for Global System for Mobile Communication. It is a digital mobile network commonly utilized by mobile phone users around the world. It is the most popular of the three digital wireless telephony systems (TDMA, GSM, and CDMA).

1. Teleservices

These are services that allow the user to **communicate directly with another user**.

Examples:

- **Voice telephony** – normal mobile voice calls
- **Emergency calls**
- **SMS (Short Message Service)**
- **Fax services**
- **Videotext**
- **Teletex**
- **Voice mailbox services**

2. Bearer Services

Bearer services provide the communication capability for carrying data between users or networks**.**

Examples include:

- Data transmission
- Circuit-switched data
- Packet data services
- Data transfer at different supported rates

3. Supplementary Services

These are **additional services provided along with basic teleservices or bearer services**.

Examples:

- **Call forwarding**
- **Call waiting**
- **Call hold**
- **Call barring**
- **Caller identification**
- **Multiparty/conference calling**
- **Advice of charge**

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12. Distinguish between HSPA,HSUPA,HSDPA

S.No.	Parameter	HSPA	HSDPA	HSUPA
1	**Full Form**	High Speed Packet Access	High Speed Downlink Packet Access	High Speed Uplink Packet Access
2	**Main Purpose**	Provides high-speed packet access for **both directions**	Improves **downlink** data transmission	Improves **uplink** data transmission
3	**Direction**	**Uplink + Downlink**	**Downlink**	**Uplink**
4	**Downlink Data Rate**	Up to about **14.4 Mbps** in early HSPA	Up to about **14.4 Mbps** in the original HSDPA; later enhancements increased this substantially	Uses the underlying HSDPA downlink; HSUPA itself does not define the downlink
5	**Uplink Data Rate**	Up to about **5.76 Mbps** in HSPA	Uses the underlying uplink technology; HSDPA itself focuses on downlink	Up to about **5.76 Mbps**
6	**Main Technology/Channel**	Combination of HSDPA and HSUPA enhancements	**HS-DSCH** (High-Speed Downlink Shared Channel)	**E-DCH** (Enhanced Dedicated Channel)
7	**Typical Application**	Web browsing, video calling, multimedia and mobile Internet	Downloading web pages, videos and files	Uploading photos, videos and files
8	**Key Point**	**HSPA = HSDPA + HSUPA**	**D = Downlink**	**U = Uplink**
